Tissular Responses in Rat Lungs Following Acute Gasoline Vapour Inhalation

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ABSTRACT

With the increasing global population and the dependence of human activities on crude oil and crude oil products, there is a resultant increase in crude production, as well as increased exposure to these products. The major route of human exposure has been via inhalation and skin contact. There are reported effects of COP exposure on organ/system, but evidence of lungs tissular response are rare. It is the objective of this study to investigate the tissular response of the lungs to acute inhalation of gasoline vapour using Wistar rats. Fifteen male Wistar rats (80–120g) were divided to 3 groups: a control and two test groups exposed to gasoline vapour for 1 and 3 hours respectively for 6 days in an exposure chamber. They were maintained under standard laboratory conditions and fed rat chow and clean water *ad libitum* throughout the experiment. Body weight was determined every other day and at the end of 6 days of exposure, blood samples were collected and the lungs harvested for weight determination, and biochemical and histological evaluations. Statistical analysis (ANOVA and LSD post-hoc) was done using SPSS (version 20) and $p<0.05$ was considered significant. The results showed acute inhalation of gasoline vapour to cause an increase in percentage weight gain, Nitric oxide level, Na^+-K^+- and H^+-K^+ pump activities, but decreased GSH and relative weight of the lungs. Histologically, the exposure to gasoline resulted in emphysematous alveoli space and desquamated airway. These results imply that exposure to acute inhalation of gasoline vapour may decrease respiratory efficiency through inhibited mucus secretion and antioxidant and stimulated nitric oxide and Na^+-K^+- ATPase activities to cause dysfunction and distorted lungs tissue histology.

Keywords: Gasoline vapour, Inhalation, Lung histology.

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1. INTRODUCTION

There have been growing industrial establishments in Nigeria and this has resulted in increased daily demand for gasoline and other petroleum products use as fuel for electricity generators at homes, offices and industries[1, 2], vehicles, cooking and lightning fuel in homes and outside homes, as chemical feedstock for industries and therapeutic reasons[3]. Gasoline which generally contains alkenes, alcohol and ether[4], is one of the fractional distillation products of crude oil known to be an inflammable, volatile[5] and a mixture of toluene, xylene, benzene, hydrocarbons and more than 500 branched and straight chains saturated, unsaturated aliphatic and aromatic compounds having 3–12 carbons[2, 6]. There have been several inclusions of additives like antiknocks and octane enhancer to gasoline to enhance its performance[7].

Gasoline is considered an environmental pollutant and toxicant that is found abundant in the atmosphere due to its volatile nature[8, 9] and has become one of the main contributors of environmental and global problems[10]. Human exposure has been reported to be through oral, skin contact and inhalation[9, 11]. The increasing industrial establishments and several applications of gasoline have increased the frequency at which individuals are exposed to its fume[2] with some reported effects on the users and greater effects among those occupationally exposed counterparts[12].

Gasoline inhalation in experimental animals has been documented to alter blood chemistry and induce anaemia by causing bone marrow hypoplasia[8], as well as reported adverse effects on some organs[7]. Other studies have reported hepatotoxicity, nephrotoxicity and alter lipids metabolisms and some biochemical activities following gasoline vapour exposure[13-15]. In another study by Owagboriaye *et al*.[16], gasoline fume inhalation was reported to increase serum lipid peroxidation, corticosterone and aldosterone level. Worrisome, there is paucity of literature on the effect of gasoline vapour inhalation on lungs tissular response even though the lungs are a major exposure route. It is the aim of this study to investigate the tissular response of the lungs to acute inhalation of gasoline vapourusing in Wistar rats.

2. MATERIALS AND METHODS

2.1. Experimental animals/ care of laboratory animals: Fifteen (15) adult male rats; Wistar strain, were obtained from the Animal Holding Unit of the College of Medicine, Ambrose Alli University. They were moved to the animal house of the Department of Anatomy, College of Medicine, Ambrose Alli University, Ekpoma for 2 weeks acclimatization. During the period of acclimatization, the rats were fed rat chow (Livestock Feed PLC, Ikeja, Lagos, Nigeria) and clean water *ad libitum* in ceramic feeding troughs and drinkers. They were housed in clean plastic cages (measuring 45cm x 35cm x 25cm) with top wire graze for ventilation under standard housing conditions of 12hour natural light/dark cycle. All the animals were given standard care and all experimental procedures conformed to the Health Guidelines for Care and Use of Laboratory Animals in Biomedical Research[17].

2.2. Animal grouping: The animals (15 male Wistar rats) were divided into 3 groups of five rats each as follows;

Group 1: Control, non exposed group.

Group 2: Exposed to gasoline (fuel) vapour for 1 hour/ day.

Group 3: Exposed to gasoline (fuel) vapour for 3hours/ day.

2.3. Gasoline (Fuel) vapour exposure: The method of exposure previously described by Owagboriaye *et al*.[2], Okonkwo *et al*.[7] and Abubakar *et al*.[18], was adopted with some modifications for this study. Gasoline (Fuel obtained from Texaco Nig. Ltd petrol station in Ekpoma) was used as the regular fuel in this study. Rats were exposed to its vapour for 1 hour (group 2) and 3 hours (group 3) per day over a period (morning) for six consecutive days. The choice of the duration of exposed in the present study was based on previous studies[2, 7, 18].

To perform the gasoline vapour inhalation, a locally fabricated rectangular inhalation chamber (90 cm x 45cm x 45cm) was used. The inhalation chamber consisted of an inlet chamber (5cm x 10cm x 5 cm; at the bottom of the breath side) demarcated with wire net from the inhalation chamber and an outlet circular channel (with diameter 10 cm; at the top side of the other breath end) where the Petri dish containing fuel was placed.

Briefly, 20 ml of fuel was introduced into a petri dish and placed in the inlet chamber and was left open for air to carry the vapour into the inhalation chamber and out through the outlet channel during the exposure period. The rats were placed to move freely in the inhalation chamber. A pilot study was conducted to estimate the time for 10ml of fuel to evaporate and it was approximately 2 hours and 25 minutes and that necessitated the used of 20 ml in the Petri dish. The chamber cover was removed after the exposure period (1hr and 3 hrs for group 2 and 3 respectively) and the animals were exposed to ambient air and returned to their housing cages for the rest of the day. This procedure was repeated for 6 consecutive days and behavioural studies carried out after 3 and 6 days of exposure. All animals in the exposure groups were exposed

to the fuel vapour at the same time between the hours of 9am and 11:30am every day while the control was allowed to move freely in inhalation chamber without gasoline vapour and kept in Laboratory II about 50 meters away.

2.4. Sample collections and analysis: Body weight was determined at commencement and every other day using digital weighing scale. The obtained body weights were used to determined percentage weight gain of each animal in each group. At the end of the experiment (24 hours after the last exposure), rats were euthanized via cervical dislocation. The lungs were harvested following standard dissecting procedures and in compliance with the Health Guidelines for Care and Use of Laboratory Animals in Biomedical Research (National Research Council, 1985). The harvested lungs were weighted and relative weight was calculated. The right lung was fixed in 10% per volume per weight of phosphate buffer for biochemical assays. The left lung was then fixed in 10% formalin for histological processing and histo-pathological evaluation.

2.5. Lungs tissue biochemical assay: The right lung was homogenized and the homogenase centrifuged at 4000 rpm at 5^0C for 10 minutes to assay for tissue malondialdehyde (MDA), mucin, nitric oxide (NO), reduced glutathione (GSH) and Na/K and H/K pumps activities spectrophotometrically. Lipid peroxidation was determined by measuring the thiobarbituric acid reactive substances (TBARS) produced during lipid peroxidation as described by Varshney and Kale[19]. The method of Beutler *et al*.[20] was followed in the estimation of the level of reduced glutathione (GSH) in lung supernatants. NO level was estimated by determining the accumulation of its stable degradation products, nitrite and nitrate following the method of Hortelano *et al*.[21]. The method of Corne[22] was adopted for the estimation of mucin level in the homogenate. The rapid colorimetric method of Amaral *et al*.[23] and Beltowski and Wójcicka[24] were respectively adopted in determining the Na/K and H/K ATPase activities.

2.6. Histological processing: The fixed liver was taken for histological processing following standard histological procedures that encompassed; fixation, dehydration, clearing, impregnation, embedding, sectioning, staining and microscopyl[25].

2.7. Statistical analysis: The data obtained were expressed as mean ± standard error of the mean (SEM) and analyzed for statistical significance using student t test and one-way ANOVA followed by LSD test where applicable; all in Statistical Package for Social Sciences (SPSS version 21). $P < 0.05$ was considered significant. Histological findings were presented in photomicrographs and results were presented in tables and chats.

3. RESULTS

Figure 1 shows the effect of short-term inhalation of gasoline vapour on percentage body weight gain. Compared with the control, percentage body weight gain was significantly higher ($p<0.05$) in the 180 minutes exposed group (1.14±1.30 g/100g unit vs 9.23±1.72 g/100g unit) by the 3rd day of exposure and in the 60 minutes exposed group by the 6th day (6.70±0.85 g/100g unit vs 12.31±1.02 g/100g unit). Figure 2 shows the effect of short-term inhalation of gasoline vapour on relative weight of the lungs. The relative weight of the lungs decrease with the degree of gasoline vapour inhalation but the difference was not statistically significant compared with the control.

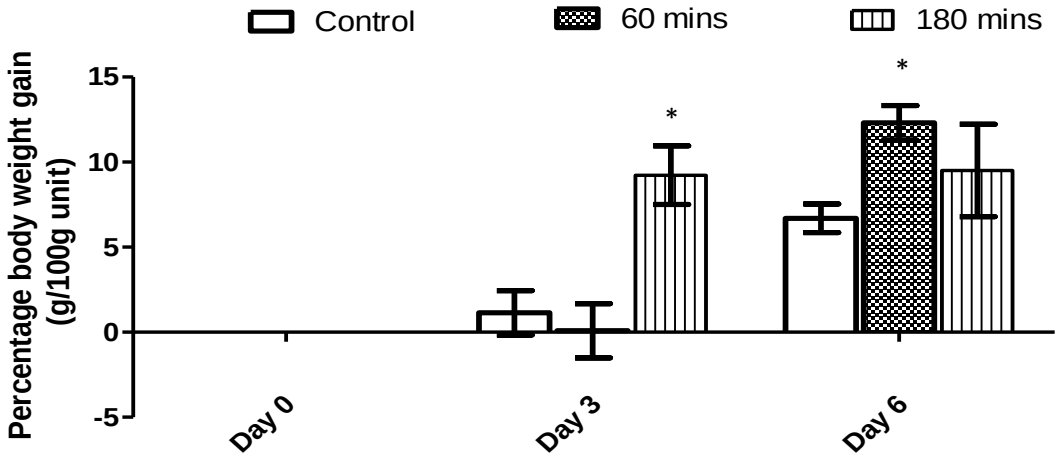


Figure 1. Effect of short-term inhalation of gasoline vapour on percentage body weight gain.
Values are in Mean ± standard error; *significantly different compared with control.

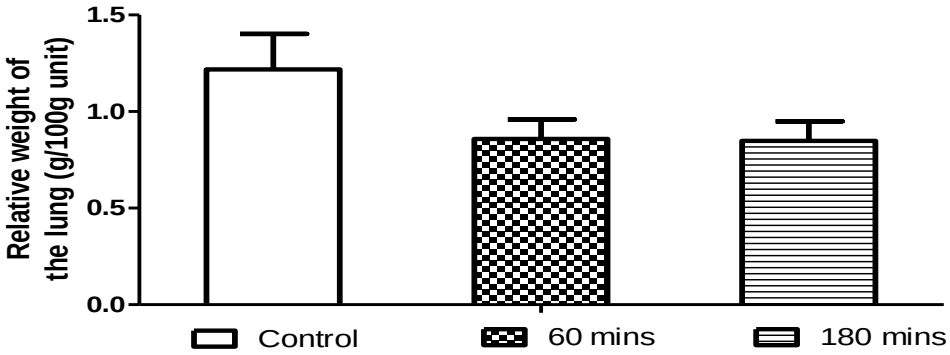


Figure 2.Effect of short-term inhalation of gasoline vapour on the relative weight of the lungs.
Values are in Mean ± standard error of mean.

Figure 3 represents the effect of acute gasoline vapour inhalation on lungs tissue malondialdehyde (MDA) level. Acute varying duration of inhalation of gasoline vapour cause a non-significant duration dependent decrease in lungs tissue MDA level compared with the control. Figure 4 represents the effect of acute gasoline vapour inhalation on lungs tissue reduced glutathione (GSH) level. Acute varying duration of inhalation of gasoline vapour cause a non-significant duration dependent decrease in lungs tissue GSH level compared with the control.

Table 1 shows the effect of acute gasoline vapour inhalation on lungs tissue secretion capacity in rats. Although not significant ($p>0.05$), acute gasoline vapour inhalation causes a duration dependent decrease in lung tissue mucin level but increase in nitric oxide level.

Table 2 shows the effect of acute gasoline vapour inhalation on lungs pump activity in rats. Acute gasoline vapour inhalation causes a dose dependent increase in lungs pump activities. Specifically, 3 hrs of gasoline vapour inhalation for 6 days significantly increase Na^+-K^+- ATPase activity compared with the control (942.30±85.55 vs 1604.50±227.84). There was no significant change in H^+-K^+ ATPase activity in the exposed groups compared with the control.

Plate 4.1 presents the histological photomicrograph of the lungs of adult male Wistar rats following acute gasoline vapour inhalation. Acute gasoline vapour inhalation for 60 and 180 minutes per day for 6 days

resulted in distortion of the normal architectural presentation of the alveoli space and airway. The observed distortion was duration dependent as only the alveoli space was affected in the 60 minutes exposed group while the alveoli space and airway were distorted in the 180 minutes exposed group.

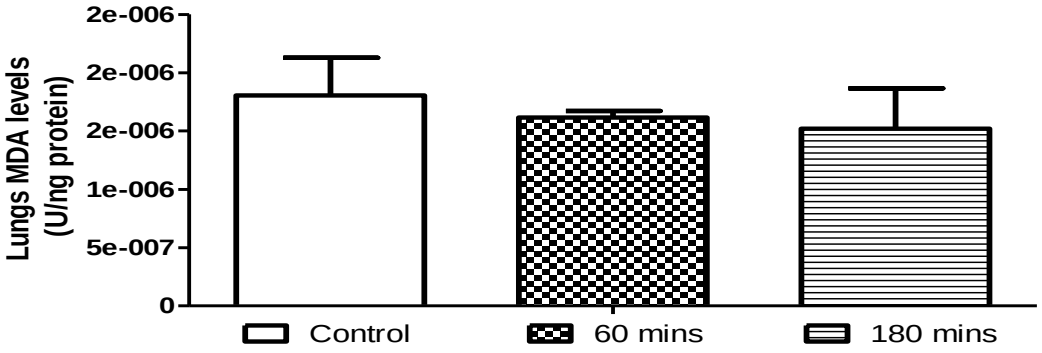


Figure 3.Effect of acute gasoline vapour inhalation on lungs tissue malondialdehyde (MDA) level.

Values are in Mean ± standard error of mean.

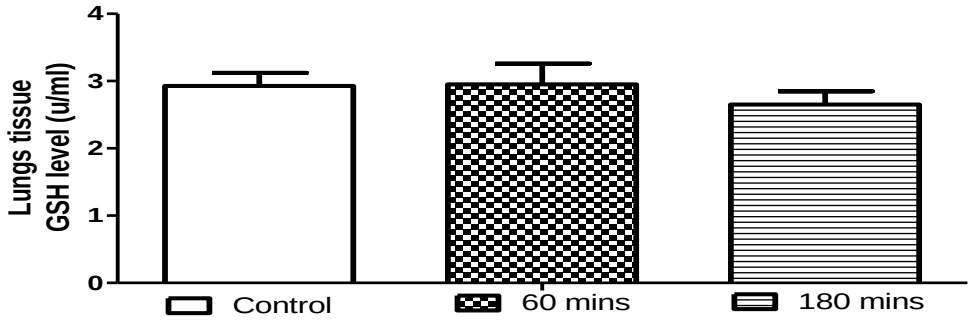


Figure 4. Effect of acute gasoline vapour inhalation on lungs tissue reduced glutathione (GSH) level.

Values are in Mean ± standard error of mean.

Table 1. The effect of acute gasoline vapourinhalation on lungs tissue secretion capacity in rats.

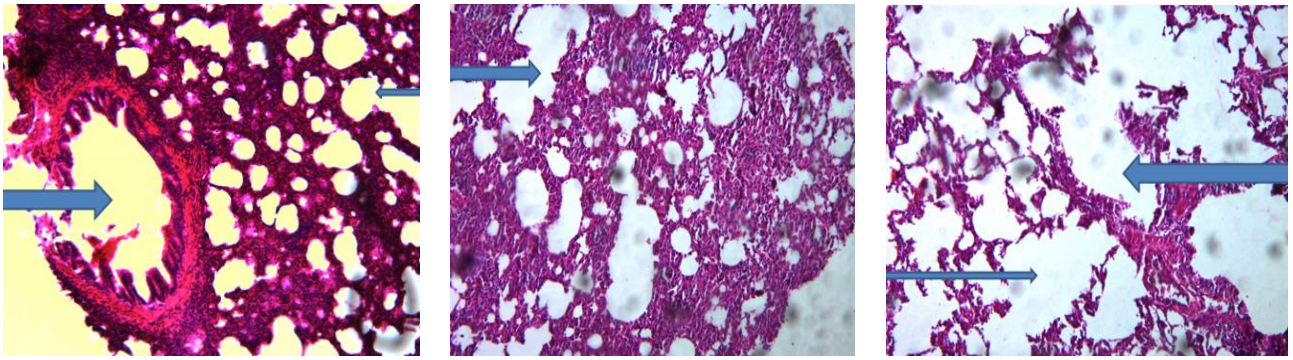
	Control	60mins fuel exposed /day	180 mins fuel exposed /day	F
Mucin	17.19±0.81	15.65±0.87	14.84±0.53	2.523
Nitric oxide	220.47±51.14	268.25±46.08	244.82±44.01	0.257

Values are in Mean ± standard error of mean.

Table 2.The effect of acute gasoline vapourinhalation on lungs pump activity in rats.

	Control	60mins fuel exposed /day	180 mins fuel exposed /day	F
Na-K ATPase	942.30±85.55	1108.50±57.74	1604.50±227.84*	5.692
H-K ATPase	1037.20±162.36	1012.30±20.75	1457.10±190.95	4.102

Values are in Mean ± standard error of mean; *significantly different compared with control.



A **B** **C**

Plate 1. Photomicrograph on the effect of short-term inhalation of gasoline vapour on the histology of the lungs. (H and E stain; Mag. X100).

A (Unexposed control) presented normal airway (thick arrow) and alveoli space (thin arrow); B (60mins fuel exposed /day) presented emphysematous alveoli space (arrow) and C (180mins fuel exposed/ day) presented desquamated airway (thick arrow) and emphysematous alveoli space (thin arrow).

4. DISCUSSION

The lungs serve the principal function of gaseous exchange. The need to balance efficient respiration in the face of tremendous exposure to environmental and endogenous challenges, the lungs maintain a sustainable defense that minimizes the damage caused by this exposure challenges and the detrimental effects of inflammation to delicate gas exchange surfaces[26]. In the present study, acute gasoline vapour inhalation in rats was observed to increase body weight gain, nitric oxide and pump activities but decrease relative weight of the lungs, tissue MDA and GSH with distortion of lung alveoli space and airways compared with the control.

Lipid peroxidation -the oxidative catabolism of polyunsaturated fatty acids, is widely reported as a general mechanism for cellular injury and death, and implicated in diverse pathological conditions[27, 28] with environmental pollutants and petrol fumes identified as factors that can enhance peroxidative processes and oxidative stress[29]. In this present study, it was found that the level of MDA was not significantly decreased compared to the control; a finding that is in disagreement with previous findings by Owagboriaye *et al*.[16], Bokolo and Ligha[30] and Uboh *et al*.[31],where it was found that gasoline exposure has the ability to increase lipid peroxidation in rats as well as the findings of Odewabi *et al*.[32] in humans.

On the other hand, the present study showed decrease GSH following acute gasoline vapour exposure. Antioxidant enzymes have been reported to act as free-radical scavengers and hence prevent and repair damage done by reactive oxygen species[33, 34]. Considering that GSH protects cells from the free radicals produced through oxidation[35], the decreased levels in the present study indicate overwhelming oxidative system.

Since the discovery that NO was similar to endothelium derived relaxing factor[36], its importance in the regulation of body functions, including the respiratory tract, has become apparent[37]. Increasing evidence has pointed to an important role for NO in the regulation of pulmonary function and in pulmonary disease[38-40]. In fact, NO has been reported to increase in various inflammatory diseases of the airways such as bronchial asthma[41]. In the present study, the lungs tissue NO level was increased in the acute gasoline vapour exposed groups compared with the control. This increased NO level in the exposed groups therefore suggest increased risk of asthma following acute gasoline vapour exposure.

The lungs are remarkably resistant to environmental injury, despite continuous exposure to pathogens, particles, and toxic chemicals in inhaled air[42]. Their resistance depends on a highly effective defense provided by airway mucus[43-46]; an extracellular gel in which water and mucin (heavily glycosylated proteins) are the most important components[42].Thus, the decrease mucin level in the lungs following acute gasoline vapour inhalation in the present study indicates decreased resistance of the lungs. This therefore suggests that acute gasoline vapour inhalation might cause a deficient mucous barrier of the lung and in turn leaves the lungs vulnerable to injury.

Na^+-K^+-ATPase pumps (Na^+ pumps) in the alveolar epithelium create a trans-epithelial Na^+ gradient crucial to keeping fluid from the pulmonary air space[47]. Na^+ pumps serve as the engine powering vectorial Na^+ transport across the alveolar epithelium and thus play an integral role in alveolar edema prevention and, if alveolar edema occurs, it then plays a role in fluid clearance[48-50]. The increased activity of Na^+ pump following acute gasoline vapour inhalation in the present study suggest that gasoline vapour inhalation may have resulted in alveolar edema and thus the activation of the pump activity to stimulate fluid clearance. A case study had strongly suggested that hydrocarbons have direct toxic effect on lung tissue, interfering with gasous exchange and leading to liquid interstitial pulmonary edema and severe hypoxemia caused by impaired diffusion[51]. Thus, the observed increase Na^+ pump following acute gasoline vapour inhalation in the present study is in accordance and may be a regulatory mechanism of the lungs in attempting to ensure fluid homeostasis in the lungs.

There has been demonstrated evidence of the presence of the H^+/K^+-ATPase (proton pump) in the lungs[52-54] and this implies a potential role of the proton pump in lungs physiology. In this study, acute inhalation of gasoline vapour was observed to stimulate the proton pump; although not to a significant level. Considering that voltage-gated H^+ channels in alveolar epithelial cells are presumed to contribute to the essential carbon dioxide (CO_2) elimination function of the lung[55], the increase stimulation of proton pump may suggest that gasoline vapour inhalation induced CO_2 retention or over production. It may therefore be that, the proton pump in the lung functions to eliminate CO_2 in the outward direction. Since facilitated diffusion in the lung does not appear to satisfy all of the physiologic requirements of CO_2 elimination, voltage-gated and other H+ channels (such as perhaps the H+/K+-ATPase proton pump) may play a crucial role[56]. DeCoursey[55] proposes that fully activated alveolar H^+ channels extrude acid equivalents at a more than sufficient rate to fulfill physiologic requirements in the lung.

The histological presentations of the lungs support the observations that gasoline vapour inhalation may inhibit mucus secretion and antioxidant level as well as induced lungs fluid imbalance and thus the observed emphysematous alveoli space and desquamated airway.

5. CONCLUSION

The results showed acute inhalation of gasoline vapour to decrease respiratory efficiency through inhibited mucus secretion and antioxidant and stimulated NO and Na-K ATPase activities to cause dysfunction and distorted lungs tissue histology. It seems unlikely that exposure to gasoline and their products will decrease in the near future considering their use in everyday life. While the acute effect has been reported in this study, long term exposures require further research and thus the recommendation of this study.

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